

Homework 6

Grant Talbert

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MA713

Boston University

Problem 1. Let X, Y be topological spaces. Prove that if X is connected then a continuous locally constant function $f : X \rightarrow Y$ is constant.

Proof. Suppose for contradiction that f is not constant. We claim for all $y \in Y$, the fiber $f^{-1}(y)$ is open. To show this, let $x \in f^{-1}(y)$. Since f is locally constant, there is an open neighborhood $x \in U_x \subseteq X$ such that f is constant at y along U_x . Then we can write

$$f^{-1}(y) = \bigcup_{x \in f^{-1}(y)} U_x,$$

which is a union of open sets, and thus open. Now note that

$$X = \bigcup_{y \in Y} f^{-1}(y)$$

is an open cover. Note also that since f is well-defined, we have that the collection $\{f^{-1}(y)\}_{y \in Y}$ is pairwise disjoint. Hence X can be realized as a union of disjoint open sets, contradicting the assumption that X be connected. ■

Problem 2. Given any open cover $\mathcal{C} = \{C_i\}_{i \in I}$ of a compact metric space (X, d) , show that there is a number $\delta > 0$ such that for every $x_0 \in X$, the ball $B_\delta(x_0)$ is contained in C_i for some $C_i \in \mathcal{C}$. (Hint: use sequential compactness)

Proof. Metric spaces are compact precisely when they are sequentially compact. Suppose for contradiction that no such δ exists. Then for all $n \in \mathbb{Z}_{\geq 1}$, there is $x_n \in X$ such that $B_{1/n}(x_n)$ is not a subset of C_i for all i ¹. Since X is compact, $\{x_n\}_{n \geq 1}$ has a convergent subsequence $\{x_{n_i}\}_{i \geq 1}$. Let $x := \lim_i x_{n_i}$. Then by definition of convergence, for all open neighborhoods $U \ni x$, there is N such that $n \geq N$ implies $x_n \in U$. ■

Problem 3. A set $S \subseteq \mathbb{R}^n$ is star-shaped if there is a point $s_0 \in S$ such that for all $s \in S$, the line segment from s_0 to s is contained in s . Prove a star-shaped set is path-connected.

Proof. Let $s, t \in S$. Then since straight lines are continuous, the paths $s_0 \rightarrow s$ and $s_0 \rightarrow t$ are continuous. By reversing the parameterization via precomposing by the homeomorphism $\varphi : I \cong I$ given by $x \mapsto 1 - x$, we obtain paths $s_0 \rightarrow t$ and $s \rightarrow s_0$. Concatenating the paths yields a path $s \rightarrow t$. ■

Problem 4. Prove that the continuous image of a path-connected space is path-connected.

Proof. Let $f : X \rightarrow Y$ with X path connected. Let $y_1, y_2 \in \text{Im } f$ and choose $x_1 \in f^{-1}(y_1)$ and $x_2 \in f^{-1}(y_2)$. Since X is path-connected, there exists a path $\gamma : I \rightarrow X$ that goes $x_1 \rightarrow x_2$. Then $(f \circ \gamma)(0) = f(x_1) = y_1$ and $(f \circ \gamma)(1) = f(x_2) = y_2$, so $f \circ \gamma : I \rightarrow Y$ is a path $y_1 \rightarrow y_2$ in $\text{Im } f$. Hence the image is path-connected. ■

¹If this is not true, then $B_{1/n}(x) \subseteq B_{1/m}(x)$ whenever $n \leq m$ implies that there is some N such that $B_{1/n}(x) \not\subseteq C_i$ for all i implies $n > N$, thus $\delta := N$ would be valid, a contradiction.